9.3. The Cartan matrix of a finite dimensional algebra

Let A be a finite dimensional algebra with simple modules M_i, $i = 1, \dots, n$, and projective covers P_i. Let $c_{ij} := \dim \mathrm{Hom}_A(P_i, P_j) = [P_j : M_i]$.

Definition 9.3.1. The matrix $C = (c_{ij})$ is called the **Cartan matrix** of A.

Obviously, the Cartan matrix of A is a matrix with nonnegative entries, whose diagonal entries are positive.

Problem 9.3.2. Let A be the algebra over complex numbers generated by elements g, x with defining relations $gx = -xg, x^2 = 0, g^2 = 1$. Find the simple modules, the indecomposable projective modules, and the Cartan matrix of A.

9.4. Homological dimension

Let A be a ring, and let M be a left A-module.

Definition 9.4.1. The **projective dimension** $\mathrm{pd}(M)$ of M is the length of the shortest finite projective resolution of M (where we agree that the length of $0 \to P_d \to \cdots \to P_0$ is d). If a finite projective resolution of M does not exist, then the projective dimension of M is infinite.

For instance, the projective modules are the modules of projective dimension 0.

Problem 9.4.2. (i) Show that $\mathrm{pd}(M) \le d$ if and only if for any left A-module N, one has $\mathrm{Ext}^i(M, N) = 0$ for $i > d$.

Hint: To prove the "if" part, use induction in d and the long exact sequence of Ext groups in Problem 8.2.6(v).

(ii) Let $0 \to M \to P \to N \to 0$ be a nonsplit short exact sequence, and assume that P is projective. Show that $\mathrm{pd}(N) = \mathrm{pd}(M) + 1$.

(iii) Show that if $\mathrm{pd}(M) = d > 0$ and $P_\bullet$ is any projective resolution of M, then the kernel K_d of the map $P_{d-1} \to P_{d-2}$ in this